LET Mathematics: Number Sense, Models, and Data Reasoning

LET · mathematics · BoardPrep lesson v1

Official examination scope

The PRC LEPT TOS and DepEd Mathematics curriculum are the scope references. This lesson teaches selected quantitative reasoning with original numbers; check the current curriculum and examination documents before treating it as a complete specialization guide.

Learning objectives

- Choose an operation and representation that fits a word problem.
- Solve and check a one-variable equation or proportional relationship.
- Summarize and compare data using the correct unit and baseline.

1. Number sense, operations, and units

Estimate before calculating so an answer with an impossible sign or scale is easy to notice. Keep the unit attached to each quantity and convert to a common unit before adding or comparing. Fractions, decimals, and percentages are alternate representations of a relationship; choose the one that makes the calculation transparent. In multi-step problems, state what each intermediate value represents instead of copying numbers into an unexplained expression. A reasonable estimate and a reverse check can reveal errors that a calculator will not.

2. Equations, patterns, and functions

Translate a verbal condition into an equation by naming the unknown and defining each term. Solve with equivalent operations, then substitute the result into the original statement. A pattern suggests a rule; verify the rule against every given term before extending it. A function describes how an output depends on an input under a defined relationship, and a graph can show where the relationship rises, falls, or changes. Distinguish a model’s prediction from an observed measurement, and state the range where the model makes sense.

3. Geometry and measurement

Geometry problems are easier when you draw and label the figure before selecting a formula. Identify the requested quantity—length, perimeter, area, angle, or volume—and check that the formula matches the shape. Use the stated units consistently and remember that area uses square units. A diagram may not be drawn to scale, so use the given measurements and relationships rather than visual appearance. For angle or shape relationships, write the rule that connects the parts and test whether the values meet the full condition.

4. Summaries, probability, and data

Before summarizing a data set, identify the population or sample, variable, and unit of observation. The mean uses all values and can move under an extreme observation; the median is the middle ordered value and may better represent a skewed set. Probability describes a defined event within a stated sample space; it is not a guarantee about one outcome. When comparing totals or rates, use the same time window and denominator. Explain what the data support and avoid extending the result to a population or cause that was not studied.

Compare time totals

Question: Reading takes 30, 40, and 35 minutes over three days; practice takes 20, 15, and 25 minutes; review takes 10, 15, and 20 minutes. Which total is largest, and by how much over the second largest?

1. Reading: 30 + 40 + 35 = 105 minutes.

2. Practice: 20 + 15 + 25 = 60; review: 10 + 15 + 20 = 45.
3. Compare the totals and subtract 105 - 60 = 45.
Answer: Reading is largest, by 45 minutes over practice.
Check: All three totals use the same unit and three-day period.

Common misconceptions

- An answer that looks reasonable needs no unit or reverse check.
- A graph’s visual scale replaces the values and labels.
- The mean and median always describe a data set equally well.

Practice patterns

- Name the unknown and units before writing an equation.
- Check a solution in the original condition.
- Compare like units, periods, and denominators.

Official scope references

Official scope: Enhanced Tables of Specifications for the Licensure Examination for Professional Teachers — Professional Regulation Commission; Professional Regulatory Board for Professional Teachers — https://prc.gov.ph/Pages/PRBv4/ProfessionalTeachersv9.htm
Official scope: PRC Guidelines on the Implementation of the PRC-CHED JMC on LEPT Alignment — Professional Regulation Commission — https://prc.gov.ph/article/guidelines-implementation-prc-ched-jmc-alignment-lept-teacher-education-curriculum

Instructional references

Instructional reference: Precalculus 2e — OpenStax, Rice University — https://openstax.org/details/books/precalculus-2e
Instructional reference: Calculus Volume 1 — OpenStax, Rice University — https://openstax.org/details/books/calculus-volume-1
Instructional reference: NIST/SEMATECH Engineering Statistics Handbook — National Institute of Standards and Technology; SEMATECH — https://www.nist.gov/programs-projects/nistsematech-engineering-statistics-handbook

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